

東京 2025 LINUX PLUMBERS CONFERENCE

TOKYO, JAPAN / DECEMBER 11-13, 2025

Generic Voting Mechanism



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History

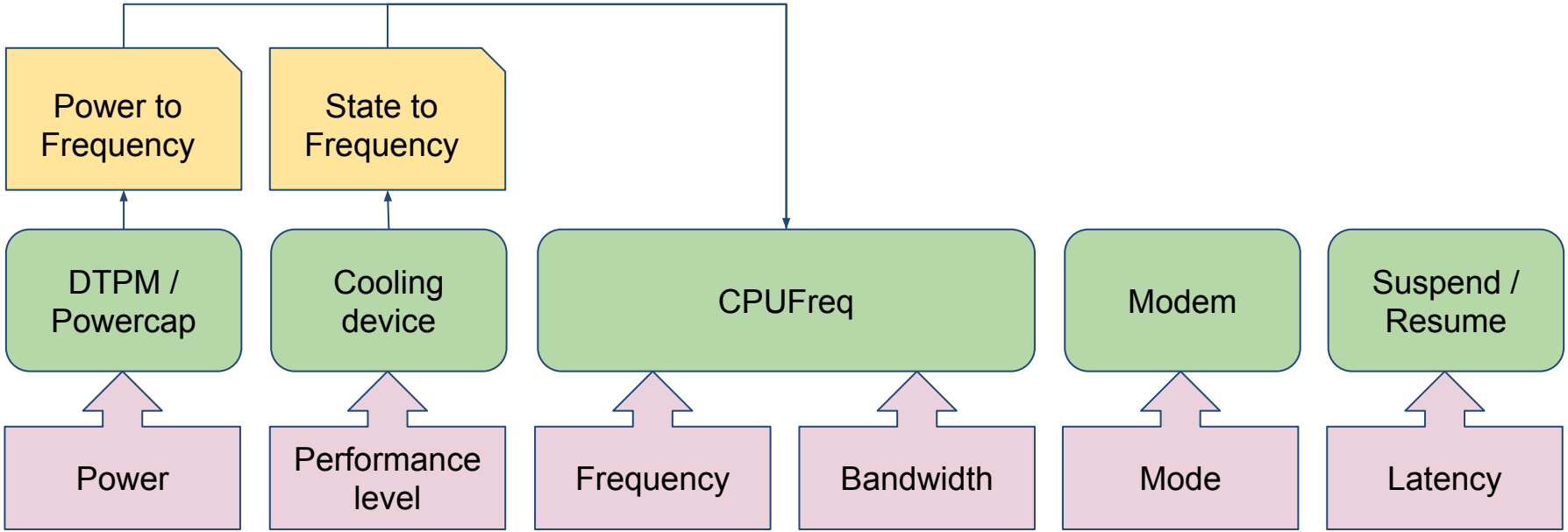
- LPC 2024
  - PM QoS framework proposal
  - Need of the userspace being able to set some constraints too
  - Consensus with char device /dev to autoremove constraints when a process closes the file descriptor (close, kill, exit)

- RFC: https://lkml.org/lkml/2025/5/5/937

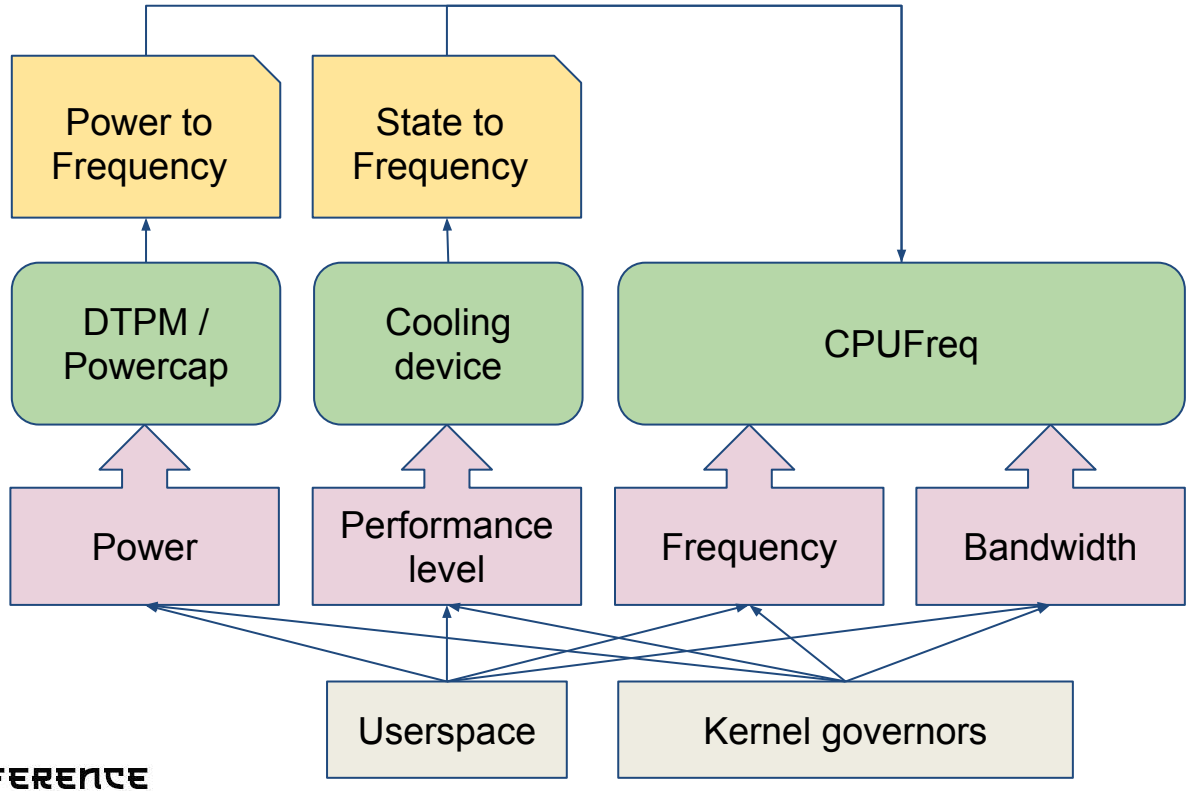
- Rafael disagrees

- It seems the confusion is coming from the used terms (constraint, performance QoS)
  - Let’s step back in the definition

Performance subsystems are connected



Multiple clients for multiple requests



Refine the needs

- It is not about performance QoS, it is about an interface to vote
- Performance level, thermal cooling state, interconnect, are voting
- Why not create a generic voting framework to be used by all the subsystems ?

The needs

- Different type of votes :
  - Auctions (Dutch or English)
  - Approval, Scoring, Weighted, …
  - Veto and Dictatorial vote (kernel only)

- Userspace and kernel can vote

- A vote happens when:
  - A voter sends a new vote
  - A voter joins or quits a ballot
  - A veto has been placed

Semantics – namespace ‘voting’

- core voting : voting_create(), voting_destroy(), voting_register(), voting_unregister(), voting_cast()
- voting policy (auctions, weighted, etc …)
  - auctions for min or max aggregation
  - weighted for trade-off performance / power decisions
- A voter can put a veto on some vote options (dependency resolution)
- A voter can force a vote with a dictatorial vote (eg. critical situation, tie-break resolution)
- Likely some algorithm taking decision can be replaced by the right voting mechanism
- But don’t overcomplicate things, let’s start with something simple : auctions

LPC 2025 - Overview

Conference Details

The Linux Plumbers Conference is the premier event for developers working at all levels of the plumbing layer and beyond.

Taking place on Thursday 11th, Friday 12th and Saturday 13th of December, this year we will be both in person and remote (hybrid). However to minimize technical issues, we'd appreciate most of the content presenters being in-person.

The in-person venue is the Toranomon Hills Forum, Tokyo, Japan

Toranomon Hills Mori Tower 5th Floor, 1-23-3 Toranomon, Minato-ku, Tokyo, 105-6305, Japan

Unless specified otherwise, the conference information will be shared in Japan (JST timezone).

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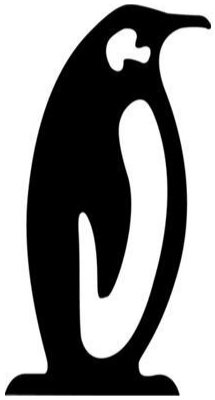
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